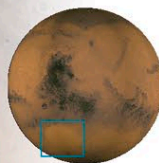


HELLAS PLANITIA

Hellas Planitia

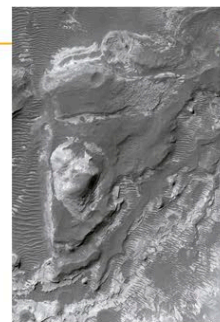
**TYPE** Basin**AGE** About 4 billion years**DIAMETER** 1,365 miles
(2,200 km)

The Hellas Basin is the largest impact crater on Mars and one of the largest in the solar system. It is the dominant surface feature in the southern hemisphere. It is not immediately apparent that Hellas is an impact crater. In fact, its official name, Hellas Planitia, indicates that it is a large, low-lying plain. This designation dates from over a century ago, when the Martian surface was observed only through Earth-based telescopes and the true nature of this vast, shallow feature was not known. Hellas is the Greek word for Greece.

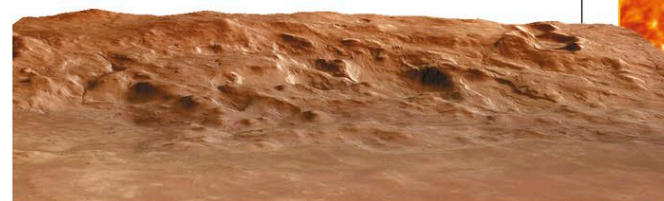
Particularly large craters that have been subsequently altered are termed basins. They are analogous to the maria on Earth's Moon. The term basin is also applied to the second-largest Martian crater, Isidis Planitia, and the third-largest,

ROCK OUTCROPS

Layered sedimentary rocks, which formed long after Hellas, lie in an eroded region northeast of the crater basin. Darker windblown ripples mark the surface.



Argyre Planitia (below). Over the past 3.5–4 billion years, Hellas Basin has had its floor filled by lava and its features changed by wind, water, and fresh crater formation. Despite all of this, some of its original features are still visible. Its overall shape and the remains of its rim can still be seen, as can inward-facing, arc-shaped cliffs lying up to several hundred miles beyond the rim. These are possibly the remnants of multiple rings.

**ERODED RIM**

This perspective view shows the northern rim—the mountain range formed around the crater as the planet's crust was lifted up at the time of impact. Whole portions of the rim are missing to the northeast and southwest.

ANCIENT BASIN

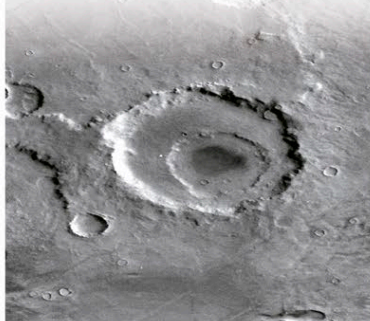
The original crater floor has been covered by volcanic and wind-borne deposits, and it also shows signs of change by water and glacial ice. Dust storms continue to shape the surface.

SIRENUM TERRA

Nansen Crater

**TYPE** Large crater**AGE** About 4 billion years**DIAMETER** 50 miles
(81 km)

Martian impact craters were first identified in 22 images returned by Mariner 4 in 1966. Nansen Crater was among the first and was named after the Norwegian explorer Fridtjof Nansen. New craters continue to be added to the list as a result of surveys by spacecraft. The Viking orbiter recorded this image of Nansen in 1976. The crater shows signs of erosion; its walls have been nibbled by the wind. Smaller, sharply defined craters have punctuated the surrounding terrain. A more recent crater has formed inside Nansen. Its central dark floor could be volcanic basalt.



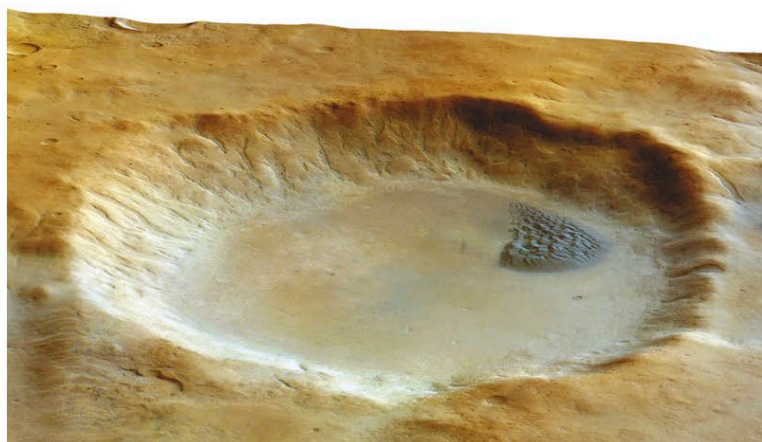
CRATER WITHIN A CRATER

ARGYRE PLANITIA

Argyre Planitia

**TYPE** Basin**AGE** About 4 billion years**DIAMETER** 500 miles
(800 km)

Argyre is the third-largest crater on Mars. Its floor has been flooded by volcanic lava, and it has been heavily eroded by wind and water. It is speculated that in the distant past, water drained into the basin from the south polar ice cap. Channels entering the basin at its southeastern edge and others leading out from its northern edge reveal the water's route. The path cuts through the mountain ranges that define the basin: the Charitum Montes to the south and the Nereidum Montes to the north.

**FROST IN THE SOUTHERN HILLS**

Frost (mainly of carbon dioxide) covers an area of cratered terrain in the Charitum Montes in early June 2003, at which time the south polar frost cap had been retreating southward for about a month.

CRATER DUNE FIELD

Argyre's floor and rugged highland rim contain smaller craters. Some of these show signs of erosion. This one lying in the northwestern part of Argyre Basin contains a dark dune field.

AONIA TERRA

Lowell Crater

**TYPE** Multiringed crater**AGE** About 4 billion years**DIAMETER** 126 miles
(203 km)

Erosion has changed Lowell since its formation early in Mars's history. The edges of both its outer rim and inner ring have been smoothed out, and its fine-grained ejecta soil has been blown around. The crater's appearance continues to undergo long-term changes, but it also changes on a short-term basis. Frost covers the crater's face in the winter months as the frost line extends north from the south polar region.



LOWELL IN WINTER